

Group 15 | Project Phase 1: Virus Breach**1. Project Overview & Vision**

Our group's arcade-style 2D game will have players take on the role of a **Virus Software** infiltrating a system server. The core experience focuses on a high-speed chase through digital corridors, where players must navigate through a grid-based map to collect data while avoiding persistent security programs out to terminate the virus. The game uses a *dynamic camera* that will follow the player's movements opposed to a static full-view map.

2. Game World & Mechanics

The game board will be a grid-based map representing a system server. The player must navigate from the "Entry Point" to the "Exit".

- **The Protagonist (The Virus):** Players will control the Virus via keyboard. The Virus moves exactly one cell up, down, left or right, to an adjacent cell per "tick". Movement is only permitted if the target cell is not obstructed by a barrier.
- **The Environment:** The server map is a maze consisting of data blocks (walls) and logic gates (individual barriers). These obstacles will block the movement of both the Virus and the Sentinels.

3. Entities & Challenges**Enemies & Obstacles:**

- **Sentinel Antivirus (Moving Enemy):** Animated antivirus programs that move one cell per tick toward the player's current position. If a Sentinel catches the Virus, the player is "deresoluted" and the game is instantly lost.
- **Firewalls & Feedback Loops (Environmental Punishment):** These are stationary hazards located on specific cells. Contact with a firewall "corrupts" the virus, leading to a score penalty.

Loot & Rewards

- **Data Packets (Regular Rewards):** Multiple packets scattered across the board. The player must collect all "regular" packets to successfully complete the mission and exit the system.
- **Source Code (Bonus Rewards):** These high-value bonus items will appear randomly during the game and disappear after a short duration. While not needed to win the game, these items offer significantly higher points to contribute to the final score.

4. Scoring and Win Conditions

- **Winning:** The player wins by collecting all required data packets and reaching the exit of the maze.
- **Losing:** The game ends in a loss if the Virus is caught by Sentinel, or the overall score drops below zero due to Firewall corruption.
- **Final Score:** The score is the accumulation of all rewards minus any penalties, which is displayed on screen along with the total playtime.